

Chapter 11.

11.1.

$$S^{-1}T = \begin{bmatrix} 0 & -\frac{1}{8} & 0 & -\frac{1}{8} & 0 & -\frac{1}{8} \\ -\frac{1}{8} & 0 & -\frac{1}{8} & 0 & -\frac{1}{8} & 0 \\ 0 & -\frac{1}{8} & 0 & -\frac{1}{8} & 0 & -\frac{1}{8} \\ -\frac{1}{8} & 0 & -\frac{1}{8} & 0 & -\frac{1}{8} & 0 \\ 0 & -\frac{1}{8} & 0 & -\frac{1}{8} & 0 & -\frac{1}{8} \\ -\frac{1}{8} & 0 & -\frac{1}{8} & 0 & -\frac{1}{8} & 0 \end{bmatrix}$$

clearly. $\|S^{-1}T\|_{\infty} = \frac{3}{8} = 0.375$

11.2.

$$A = \begin{bmatrix} 16 & 2 \\ 2 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 10 \\ -1 \end{bmatrix} \quad x = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

Jacobi method:

$$S = \begin{bmatrix} 16 & \\ & 1 \end{bmatrix} \quad T = \begin{bmatrix} 0 & -2 \\ -2 & 0 \end{bmatrix}$$

$$\begin{cases} x_1^{new} = (10 - 2x_2^{old})/16 \\ x_2^{new} = -1 - 2x_1^{old} \end{cases}$$

$$x_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} (10 - 2 \cdot 0)/16 \\ -1 - 2 \cdot 0 \end{bmatrix} = \begin{bmatrix} \frac{5}{8} \\ -1 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} (10 - 2(-1))/16 \\ -1 - 2 \cdot \frac{5}{8} \end{bmatrix} = \begin{bmatrix} \frac{3}{4} \\ -\frac{9}{4} \end{bmatrix}$$

$$x_3 = \begin{bmatrix} (10 - 2(-\frac{9}{4}))/16 \\ -1 - 2 \cdot \frac{3}{4} \end{bmatrix} = \begin{bmatrix} \frac{29}{32} \\ -\frac{5}{2} \end{bmatrix}$$

$$x_4 = \begin{bmatrix} (10 - 2(-\frac{5}{2}))/16 \\ -1 - 2 \cdot \frac{29}{32} \end{bmatrix} = \begin{bmatrix} \frac{15}{16} \\ -2\frac{13}{16} \end{bmatrix}$$

$$\|x - x_4\|_{\infty} = \frac{3}{16}$$

Gauss-Seidel method.

$$S = \begin{bmatrix} 16 & \\ 2 & 1 \end{bmatrix} \quad T = \begin{bmatrix} 0 & -2 \\ 0 & 0 \end{bmatrix}$$

$$\begin{cases} x_1^{new} = (10 - 2x_2^{old})/16 \\ x_2^{new} = -1 - 2x_1^{new} \end{cases}$$

$$x_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} (10 - 2 \cdot 0)/16 \\ -1 - 2 \cdot \frac{5}{8} \end{bmatrix} = \begin{bmatrix} \frac{5}{8} \\ -\frac{9}{4} \end{bmatrix}$$

$$x_2 = \begin{bmatrix} (10 - 2(-\frac{9}{4}))/16 \\ -1 - 2 \cdot \frac{29}{32} \end{bmatrix} = \begin{bmatrix} \frac{29}{32} \\ -2\frac{13}{16} \end{bmatrix}$$

$$x_3 = \begin{bmatrix} (10 - 2(-2\frac{13}{16}))/16 \\ -1 - 2 \cdot \frac{125}{128} \end{bmatrix} = \begin{bmatrix} \frac{125}{128} \\ -2\frac{61}{64} \end{bmatrix}$$

$$x_4 = \begin{bmatrix} (10 - 2(-2\frac{61}{64}))/16 \\ -1 - 2 \cdot \frac{509}{512} \end{bmatrix} = \begin{bmatrix} \frac{509}{512} \\ -2\frac{253}{256} \end{bmatrix}$$

$$\|x - x_4\|_{\infty} = \frac{3}{256}$$

11.3. $A = \begin{bmatrix} 4 & 1 & -1 \\ 1 & 6 & 2 \\ 1 & 1 & -5 \end{bmatrix}$ $b = \begin{bmatrix} 2 \\ -3 \\ -5 \end{bmatrix}$ $x = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

Jacobi method.

$$S = \begin{bmatrix} 4 & & \\ & 6 & \\ & & -5 \end{bmatrix} \quad T = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 0 & -2 \\ -1 & -1 & 0 \end{bmatrix}$$

$$x_1^{\text{new}} = (2 - x_2^{\text{old}} + x_3^{\text{old}})/4$$

$$x_2^{\text{new}} = (-3 - x_1^{\text{old}} - 2x_3^{\text{old}})/6$$

$$x_3^{\text{new}} = (-5 - x_1^{\text{old}} - x_2^{\text{old}})/(-5)$$

$$x_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} (2 - 0 + 0)/4 \\ (-3 - 0 - 0)/6 \\ (-5 - 0 - 0)/(-5) \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} (2 - (-\frac{1}{2}) + 1)/4 \\ (-3 - \frac{1}{2} - 2)/6 \\ (-5 - \frac{1}{2} + \frac{1}{2})/(-5) \end{bmatrix} = \begin{bmatrix} \frac{7}{8} \\ -\frac{11}{12} \\ 1 \end{bmatrix}$$

$$x_3 = \begin{bmatrix} (2 + \frac{11}{12} + 1)/4 \\ (-3 - \frac{7}{8} - 2)/6 \\ (-5 - \frac{7}{8} + \frac{11}{12})/(-5) \end{bmatrix} = \begin{bmatrix} \frac{47}{48} \\ -\frac{47}{48} \\ \frac{119}{120} \end{bmatrix}$$

$$x_4 \approx \begin{bmatrix} 0.9927 \\ -0.9938 \\ 1.000 \end{bmatrix}$$

$$\|x - x_4\|_{\infty} \approx 0.0073$$

Gauss-Seidel method.

$$S = \begin{bmatrix} 4 & & \\ 1 & 6 & \\ 1 & 1 & -5 \end{bmatrix} \quad T = \begin{bmatrix} 0 & -1 & 1 \\ 0 & 0 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1^{\text{new}} = (2 - x_2^{\text{old}} + x_3^{\text{old}})/4$$

$$x_2^{\text{new}} = (-3 - x_1^{\text{new}} - 2x_3^{\text{old}})/6$$

$$x_3^{\text{new}} = (-5 - x_1^{\text{new}} - x_2^{\text{new}})/(-5)$$

$$x_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} (2 - 0 + 0)/4 \\ (-3 - \frac{1}{2} - 0)/6 \\ (-5 - \frac{1}{2} + \frac{1}{2})/(-5) \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ -\frac{7}{12} \\ \frac{59}{60} \end{bmatrix}$$

$$x_2 = \begin{bmatrix} (2 + \frac{7}{12} + \frac{59}{60})/4 \\ (-3 - \frac{107}{120} - \frac{59}{36})/6 \\ (-5 - \frac{107}{120} + \frac{703}{120})/(-5) \end{bmatrix} = \begin{bmatrix} \frac{107}{120} \\ -\frac{703}{720} \\ \frac{3539}{3600} \end{bmatrix}$$

$$x_3 \approx \begin{bmatrix} 0.9899 \\ -0.9927 \\ 0.9994 \end{bmatrix}$$

$$x_4 \approx \begin{bmatrix} 0.9980 \\ -0.9995 \\ 0.9997 \end{bmatrix}$$

$$\|x - x_4\|_{\infty} \approx 0.0020$$

11.4. Yes. Since.

$$S = \begin{bmatrix} 2 & & \\ -6 & 9 & \\ 8 & -8 & 8 \end{bmatrix} \quad T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$S^{-1}T = \begin{bmatrix} 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{3} & \frac{2}{9} \\ 0 & \frac{1}{4} & \frac{1}{2} \end{bmatrix} \Rightarrow \|S^{-1}T\|_{\infty} = 0.75 < 1.$$

11.5. Trivial proof.

11.6. $D = \begin{bmatrix} 2 & & \\ & 3 & \\ & & 2 \end{bmatrix} \quad L = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad U = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad X = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$

$$X^{new} = X^{old} + w \cdot \begin{bmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{2} \end{bmatrix} \left(\begin{bmatrix} 7 \\ 8 \\ 5 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X_1^{new} \\ X_2^{new} \\ X_3^{new} \end{bmatrix} - \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} X_1^{old} \\ X_2^{old} \\ X_3^{old} \end{bmatrix} \right)$$

$$X_1^{new} = X_1^{old} + \frac{w}{2} [7 - 2X_1^{old} - X_2^{old}]$$

$$X_2^{new} = X_2^{old} + \frac{w}{3} [8 - X_1^{new} - 3X_2^{old} - X_3^{old}]$$

$$X_3^{new} = X_3^{old} + \frac{w}{2} [5 - X_2^{new} - 2X_3^{old}]$$

Let $X^0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, $w=1$, we have

$$X_1^1 = \begin{bmatrix} 3.85 \\ 1.5217 \\ 1.9131 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 2.6281 \\ 1.1161 \\ 1.9449 \end{bmatrix}, \quad X_3 = \begin{bmatrix} 2.9734 \\ 1.0184 \\ 1.9954 \end{bmatrix}, \quad X_4 = \begin{bmatrix} 2.9926 \\ 1.0026 \\ 1.9990 \end{bmatrix}$$

$$\|X - X_4\|_{\infty} = 0.0074$$

Let $X_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ and $w=1$, we have.

$$X_1 = \begin{bmatrix} 3.5 \\ 1.5 \\ 1.75 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 2.75 \\ 1.1667 \\ 1.9167 \end{bmatrix}, \quad X_3 = \begin{bmatrix} 2.9167 \\ 1.0556 \\ 1.9722 \end{bmatrix}, \quad X_4 = \begin{bmatrix} 2.9722 \\ 1.0185 \\ 1.9907 \end{bmatrix}$$

$$\|X - X_4\|_{\infty} = 0.0278$$

11.7. Proof: $SX^{new} = b + TX^{old} \Rightarrow X^{new} = S^{-1}b + S^{-1}(S-A)X^{old}$
 $\Rightarrow X^{new} = S^{-1}b + X^{old} - S^{-1}AX^{old}$
 $= X^{old} + S^{-1}(b - AX^{old})$